

HUMAIRA ALI

Early-career Digital Specialist Engineer pursuing a Bachelor of Technology in Electronics and Communication Engineering

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in embedded systems, data analysis, AI/ML-enabled applications, and product development. Experienced in building ESP32-based intelligent systems, analyzing sensor data, developing automation solutions, and collaborating on customer-focused products. Passionate about Product Analytics, AI-driven decision making, and building technology solutions that create measurable business impact.

SKILLS

Programming :	python, SQL, Verilog, C, C++, Java, HTML
Data & Analytics :	Data Analysis, Data Visualization, Microsoft Excel, Power BI, Pandas, NumPy, SQLite, MySQL
AI & Machine Learning :	Machine Learning, Artificial Intelligence, Predictive Analytics, Data Preprocessing, Prompt Engineering, Generative AI, OpenAI ChatGPT, Google Gemini, Perplexity, Claude
Soft Skills :	Communication, Leadership, Teamwork, Problem Solving, Critical Thinking, Adaptability, Time Management, Collaboration, Decision Making
Tools & Platforms :	MATLAB, Keil uVision, LTspice, Proteus, SQLite, GitHub, Git, Linux, Power BI, Excel, VS Code, Jupiter Notebook
Product & Business :	Product Analysis, Root Cause Analysis, Product Lifecycle, Stakeholder Communication
Languages :	English, Hindi, Arabic, Urdu

PROJECTS

Smart Helmet System for Accident Detection & Navigation

(3-Members)

- Designed and developed an ESP32-based intelligent safety system to address the high rate of two-wheeler accidents in India by integrating multi-sensor data, real-time accident detection, Bluetooth communication, and GPS-enabled emergency alerts. Analyzed sensor data and optimized detection thresholds to improve accuracy while minimizing false emergency notifications.
- Designed and developed an **ESP32-based smart helmet** for real-time accident detection and emergency response to improve rider safety.
- Integrated **MPU6050 and ultrasonic sensors** to enable crash detection, rear-obstacle awareness, and rider assistance using a multi-sensor embedded system.
- Implemented a **3G threshold-based accident detection algorithm** with a **10-second user cancellation window**, Bluetooth communication, and GPS-enabled SOS alerts to minimize false emergency notifications.
- Leveraged **AI/ML research** to analyze accident patterns, evaluate detection techniques, and optimize the system for improved reliability and real-world performance. Included **Analog Layout Concepts** and **microcontrollers** in the design process.

Technologies / Tools Used : ESP32, Embedded C++, Bluetooth, MPU6050, Ultrasonic Sensor, Android, Artificial Intelligence, Machine Learning, microcontrollers

Morse Code Encoder-Decoder Device

Designer

- Designed a bidirectional **Arduino-based Morse code encoder and decoder** supporting configurable transmission speeds from **100-600 ms**.
- Implemented dual communication modes, LCD-based decoding, adaptive timing control, SOS signaling, and configurable encoding logic.

Technologies / Tools Used : Arduino, Embedded C++, LCD, Signal Encoding

RFID-Based Multi-Module Campus Management System

Developer

- Developed an **ESP32- and RFID-based campus management platform** integrating attendance, library management, and cafeteria billing across 3 operational modules.
- Automated attendance logging, PDF bill generation, Excel exports, and database management through a **Python GUI application**.

Technologies / Tools Used : ESP32, Python, SQLite, RFID, Bluetooth, Tkinter

WORK EXPERIENCES

Intern | Internship

Bharat Heavy Electricals Limited (BHEL)

May 2026 - Jun 2026

Bhopal, Madhya Pradesh

Developed PLC-based industrial automation solutions to improve operational efficiency. Collaborated with engineers to analyze manufacturing workflows, troubleshoot control systems, and support embedded firmware development. Contributed to process optimization through automation and real-time monitoring.

Technologies / Skills Used : Semiconductor Fundamentals, Embedded Firmware Development

CERTIFICATIONS

- Introduction to Programming in C** — IIT Kharagpur
- Internet of Things** — IIT Kharagpur
- MATLAB Signal Processing** — MathWorks
- VLSI Design & Verification Program** — Ethnus
- Google IT Support Professional Certificate** — Google
- Introduction to Self-Driving Cars** — University of Toronto
- Bits and Bytes of Computer Networking** — Google

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Database Management Systems
- Operating Systems
- Computer Networks
- Machine Learning & Artificial Intelligence
- Embedded Systems
- Digital Electronics
- Microprocessors
- Software Engineering
- VLSI
- Signals and Systems

EDUCATIONS

Vellore Institute Of Technology

Bachelor of Technology in Electronics and Communication Engineering

- 10th: **89.8/100**, Carmel Convent BHEL-CBSE, 2021
- 12th: **80.4/100**, Carmel Convent BHEL-CBSE, 2023

2023 - 2027

CGPA : 8.66 / 10